TT-P-91D
February 26, 1974
SUPERSEDING
Int. Fed. Spec. TT-P-0091C
July 27, 1971
INTERIM REVISION OF
Fed. Spec. TT-P-91a
April 10, 1950

FEDERAL SPECIFICATION

PAINT, RUBBER-BASE, FOR INTERIOR USE (CONCRETE AND MASONRY FLOORS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a rubber-base paint for interior use on concrete and masonry floors. When this specification is specified without reference to Type, either Type may be supplied. Both types have the same end use.

1.2 Type. The rubber-base paint shall be of the following:

Type I - Chlorinated Rubber.
Type II - Styrene-butadiene.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-B-531 - Brush, scrub; floor, hand.
O-S-642 - Sodium Phosphate, Tribasic, Technical; Anhydrous, Dodecahydrate, and Monohydrate.
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of.

Federal Standards:

Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials; Method of Inspection, Sampling, and Testing.
Federal Standard No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 Paint. The paint, as received, shall be ready-mixed for use and of brushing consistency. It shall consist of pigment and vehicle specified, so combined as to produce a paint meeting all the requirements of this specification.

3.2 Pigment. Any suitable pigments including coloring pigments may be used providing the paint meets all the requirements of this specification.

3.3 Vehicle. The vehicle shall consist essentially of a natural-rubber base (chlorinated rubber) for type I and styrene-butadiene (synthetic) for type II. Suitable plasticizers for each type shall be used provided the product complies with all the requirements. The solvent shall be mineral spirits conforming to TT-T-291, type II or any solvent system which complies with "Air Pollution Regulations" Rule 66 1/.

3.4 Quantitative requirements. Rubber-base paints shall meet the quantitative requirements specified in table I.

Table I. Quantitative requirements of rubber-base paint

Characteristics	Requirements	
	Minimum	Maximum
Rubber-base precipitate, percent by weight of vehicle	20	---
Water, percent by weight of paint	---	1
Drying time, hours:		
Set-to-touch	1/4	3/4
Hard	---	3
Specular gloss, 60°	75	---
Consistency, Krebs-Stormer, shearing rate 200 r.p.m.:		
Grams	125	225
Equivalent	67	86
Abrasion resistance (Wear Index)	---	60
Hiding power (contrast ratio) *Note	0.98	---

*NOTE: Applied on a white and black carrara glass panel at a maximum dry film thickness of 1 mil.

3.5 Qualitative requirements.

3.5.1 Condition in container. The paint, when tested as in 4.3.2, shall be well-ground and not caked, livered, thickened, skinned, nor badly settled in the container. Any settling shall be capable of being easily broken up and redispersed in the liquid portion to form a smooth paint of good brushing consistency.

3.5.2 Color. When tested as in 4.3.3 the color shall be a match of the reference standard of Fed. Std. No. 595 as specified by the procuring agency. (see 6.2 and 6.3).

3.5.3 Application properties. The paint, when tested as in 4.3.4, shall show easy brushing, good flowing, and good leveling properties. The paint shall not pull nor have a quick set under the brush. The dried film shall show good leveling and there shall be no brush marks.

3.5.4 Flexibility. The paint shall show no evidence of cracking when subjected to the flexibility test when tested as in 4.3.5.

3.5.5 Storage stability.

3.5.5.1 Partially full container. A three-quarter filled, closed 8-ounce glass jar of paint shall show no skinning when tested as in 4.3.6.1. After aging as in 4.3.6.1, the paint shall show no livering, curdling, seeding, hard caking, or gummy sediment. It shall mix readily to a smooth homogeneous state and any skin formed shall be continuous and easily removed.

3.5.5.2 Full container. A full quart container of paint shall show no skinning, livering, curdling, seeding, hard, dry caking nor tough, gummy sediment when tested as in 4.3.6.2. The paint shall remix readily to a smooth homogeneous state, shall have a maximum viscosity of 95 K.U. and shall meet all other requirements of the specification.

1/ Information on "Rule 66" may be obtained from Los Angeles Air Pollution Control District, Los Angeles, California 90013

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3.5.6 Dilution stability. When thinned as in 4.3.7, the paint shall remain stable and uniform showing no precipitation, curdling, or separation.

3.5.7 Cement-water. When tested as in 4.3.8, the paint shall show no discoloration, blistering, cracking, or flaking.

3.5.8 Detergent resistance. The paint film, when tested as in 4.3.9, shall show no evidence of detrimental action by trisodium phosphate solution or brushing action.

3.5.9 Self-lifting properties. Recoating the test surface as specified in 4.3.10 shall produce no lifting or other film irregularity.

3.5.10 Streaking. A film of the thoroughly mixed paint, flowed on a glass panel and dried in a nearly vertical position, shall show no streaking or separation of any of the paint components when tested as in 4.3.11.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize hiw own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with section 1000 of Fed. Test Method Std. No. 141

4.3 Test Procedures.

4.3.1 Test methods shall be in accordance with Fed. Test Method Std. No. 141 except as otherwise specified (see Table II).

TABLE II. Test methods

Characteristics	Requirement Reference	Test Method Fed. Test Method Std. No. 141	Paragraph Reference
Condition in container	3.5.1	3011.1	4.3.2
Color	3.5.2	4250	4.3.3
Application properties	3.5.3		4.3.4
Flexibility	3.5.4	6221	4.3.5
Storage stability	3.5.5	3021, 3022	4.3.6
Dilution stability	3.5.6	4203.1	4.3.7
Cement-water test	3.5.7	----	4.3.8 1/
Detergent resistance	3.5.8	----	4.3.9
Self-lifting properties	3.5.9	----	4.3.10
Streaking resistance	3.5.10	----	4.3.11
Rubber-base precipitate	Table I	----	4.3.12
Consistency	Table I	4281	----
Water content	Table I	4081	----
Drying time	Table I	4061.1	----
Abrasion resistance	Table I	----	4.3.13
Hiding power (contrast ratio)	Table I	4122.1	----
Gloss, 60°	Table I	6101	----

1/ Pour 80 milliliters of paint into a 100 milliliter graduated cylinder and gradually add, with stirring, a thinner or thinner system indicated in paragraph 3.3 until the 100 milliliter graduation is reached. Settling of pigment is disregarded in determining homogeneity.

4.3.2 Condition in container. Examine the paint as received in accordance with method 3011.1 of Fed. Test Method Std. No. 141 for compliance with 3.5.1.

4.3.3 Color. Apply a film of the paint to a clean, smooth, plate glass panel with a 0.003-inch (approximately 0.006-inch gap clearance), Bird film applicator or any doctor blade which produces a film of the same thickness as that produced by the Bird blade may be used. Allow to dry for 24 hours at room temperature then evaluate color in accordance with method 4250 of Fed. Test Method Std. No. 141 for compliance with 3.5.2.

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4.3.4 Application properties. Concrete panel shall be prepared in accordance with method 2051, procedure B of Fed. Test Method Std. No. 141. Remove the glaze of the troweled surface by sandpaper or wire brush. Brush a coat of the paint on the prepared surface at spreading rate of approximately 400 square feet per gallon, while brushing observe for pull and quick set under brush. Allow to dry in a horizontal position for 48 hours at room temperature, then evaluate for compliance with 3.5.3.

4.3.5 Flexibility. Draw down a film of the paint on a cleaned electrolytic tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141 using 0.002-inch blade. Air dry for 24 hours, bake for 24 hours at 105° ± 2°C. Cool at 23° ± 0.5°C. Then bend over 1/4-inch mandrel. Observe for compliance with 3.5.4.

4.3.6 Storage stability.

4.3.6.1 Partially full container. Determine skinning after 48 hours in accordance with method 3021 of Fed. Test Method Std. No. 141. Reseal and age for 7 days at 60°C. and observe for compliance with 3.5.5.1.

4.3.6.2 Full container. In accordance with method 3022 of Fed. Test Method Std. No. 141 allow a full standard quart can of the paint to stand undisturbed for 12 months and then examine the contents. Evaluate pigment settling or caking as in 4.3.2 but agitate the can for 5 minutes on the paint shaker prior to re-examination. Determine viscosity and make other applicable tests to determine compliance with 3.5.5.2.

4.3.7 Dilution stability. Determine dilution stability in accordance with method 4203.1 of Fed. Test Method Std. No. 141. Pour 80 ml. of paint into a 100 ml graduate and gradually add with stirring a thinner or solvent specified in 3.3 until the 100 ml graduation is reached. Observe in accordance with method 4203 and evaluate for compliance with 3.5.6.

4.3.8 Cement-water test. Cement blocks 1/ shall be prepared from a mix of 1 part of portland cement and 1 part of fine sand, by volume. Moisten the mixture with water and cast into blocks measuring 3 inches long by 3 inches wide by 1-1/2 inches thick. The top surface of these blocks shall be troweled smooth. Allow blocks to harden in air for at least 2 weeks (note 1). Remove the glaze with No. 1 sandpaper. Brush a coat of the paint, which has been thinned as prescribed in footnote 1 of 4.3.1, over the troweled sides of two cement blocks and allow to dry for 24 hours. Apply a second coat, using the paint, as received, at an approximate spreading rate of 400 square feet per gallon. Allow the second coat to dry for 48 hours. Then immerse the blocks in water to such depths as to have the prepared surface 1/4-inch above the surface of the water. Allow the blocks to remain partially immersed for 10 days; then totally immerse them in water for 48 hours. Remove after this period and examine for compliance with 3.5.7.

4.3.9 Detergent test.

4.3.9.1 Trisodium phosphate. Immerse one of the blocks from the cement-water test (4.3.8) in a solution (20 grams to 2 gallons of water) of trisodium phosphate conforming to O-S-642. Maintain the solution at 50° to 60°C. While immersed, scrub the surface with a floor scrubbing brush conforming to H-B-531. A 3-pound weight shall be placed on the brush. Apply 500 strokes to the painted surface of each block. A stroke shall be considered a complete forward and reverse movement through a distance of 2 to 3 inches in each direction. The block shall then be removed. The painted surface shall be rinsed with water and wiped dry. The painted surface shall then be examined and compared with the painted surface of a block (used for comparison) subjected to the cement-water test but which has not been subjected to the detergent test. Observe whether the paint surface of the block which has been subjected to the detergent test presents the same surface appearance as the "comparison" block, without showing any signs of detrimental action by the trisodium phosphate solution or the brushing action. Observe for compliance with 3.5.8.

1/ Blocks may be prepared in advance and held ready for use; however, blocks which have aged for more than 6 months should not be used for this test.

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4.3.10 Self-lifting. Determine self-lifting properties of the paint in accordance with method 6252 of Fed. Test Method Std. No. 141. Use test panel in 4.3.4. Brush a coat at approximately 400 square feet per gallon on dried film and allow to dry at room temperature for 48 hours. Examine test film for compliance with 3.5.9.

4.3.11 Streaking. Flow a portion of the thoroughly mixed paint on a clean glass panel about 4 inches by 6 inches in size. Place the panel, resting on its shorter dimension, in a nearly vertical position to drain and dry. The test shall be made at room temperature, 21° to 32°C. (70° to 90°F.). Examine the film for streaking or separation of its components within a distance of 4 inches from the top for compliance with 3.5.10.

4.3.12 Rubber-base precipitate.

4.3.12.1 Type I (Chlorinated rubber). Determine rubber-base precipitate in accordance with procedure 4.1, method 5211.1 of Fed. Test Method Std. No. 141 for compliance with Table I. In case of noncompliance run test using Hercules (Coatings Service Laboratory) CSL-217 method No. II. 1/

4.3.12.2 Type II (Styrene-butadiene). Determine rubber-base precipitate in accordance with method 5212.1 of Fed. Test Method Std. No. 141 for compliance with Table I.

4.3.13 Abrasion resistance. Determine abrasion resistance of the paint in accordance with method 6192 of Fed. Test Method Std. No. 141 for compliance with Table I. A coat of the thoroughly mixed paint is sprayed (method 2131 of Fed. Test Method Std. No. 141) on duplicate steel panels, 4 by 4 inch, until the thickness of the test film is approximately 15 mils. After air drying at room temperature for two weeks run the test using Taber Abraser, Model 174, CS-17 calibrase Wheel, 250 grams weight, 100 revolutions, then observe for compliance with Table I.

4.3.14 Inspection of preparation for delivery. The paint shall be examined for compliance with packaging, packing, and marking requirements of section 5 in accordance with TT-P-143. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number of the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be level A, B, or C and the level of packing shall be level A, B, or C as specified (see 6.2). The paint shall be furnished in the size container specified (see 6.2).

5.1.1 Additional marking. In addition to the marking requirements specified in 5.1, the directions for use, which shall be clearly legible, shall be shown on the reverse side of each container and shall read as follows:

"This rubber-base paint is intended for painting indoor concrete floors, especially basement floors subject to dampness, but it can also be used on interior masonry walls. It is not intended for exterior use or for application over other types of paint.

For the best results, this paint should be applied directly to clean, bare, and dry concrete. New floors should age for at least 2 months before painting. Any grease or oil spots should be washed with a strong alkaline cleanser and rinsed. Floors with hard slick surfaces or those with dusting surfaces should be acid etched by flooding (1 gallon to 100 square feet) with a mixture of 1 gallon of muriatic acid added to 4 gallons of water. After 15 minutes, hose off the acid using plenty of water. Allow the floor to dry before painting. At least two coats should be used, and for extreme wear resistance three coats are recommended. The first coat should be thin, 1 part thinner (Thinner, see 3.3) to 4 parts paint (by volume). The second and third coats may be applied as received. The spreading rate per gallon of paint is approximately 300 to 400 square feet per coat. While the paint dries tack-free within 1 hour, overnight drying is recommended between coats. Provide ventilation while applying the paint. The paint may be applied by brushing or spraying (add 1 pint of thinner to each gallon paint for spraying).

The paint should be stored under good conditions (40° to 90°F.) and the estimated "shelf life" (closed containers) is 2 years.

1/ This test methodmmay be obtained from Hercules Incorporated, 910 Market Street, Wilmington, Delaware 19899.

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Caution: Keep paint away from flames. Provide adequate ventilation while applying the paint. Avoid prolonged inhalation of vapors."

6. NOTES

6.1 Intended use. This specification covers a ready-mixed rubber-base floor paint intended for use on interior concrete floors subject to dampness. It is not intended for exterior use or for exposure to sunlight.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable level of packaging and packing required (see 5.1).
(d) Color required (see 3.5.2).
(e) Quantity and size of container required (see 5.1).

6.3 The paint covered by this specification can be satisfactorily thinned when necessary, with the thinner specified in footnote 1 of 4.3.1. On unpainted concrete, three coats are recommended. On smooth, densely troweled concrete floors, it is recommended that the surface be etched with dilute (10 percent solution) muriatic (hydrochloric) acid, washed, and the floor allowed to dry thoroughly before applying this paint. New concrete floors should be allowed to age for at least 2 months before painting with this paint. On repainting, it is advisable to use this type of paint only over the same type of paint. As in case of other types of floor paint, this paint should be applied to a dry, clean surface. Care should be taken to have the room properly ventilated when applying the paint.

6.4 Rubber-base floor paint under this specification should be purchased by volume. The unit being one U.S. liquid of 231 cubic inches at 68°F. °(20°C.).

6.5 Rubber-base paints under this specification usually dry tack-free in less than an hour, and dry sufficiently hard in 3 hours to be called "hard dry" according to the specification. Painted concrete floors can be walked after drying overnight, but it is well to allow a drying time of 48 hours before subjecting the floor to severe wear. This type of paint has good resistance to water, chemicals (dilute acids, alkalies, soap solutions) and various mineral oils and grease (such as are found on garage floors). The paint shows good resistance to abrasion and scrubbing. While its main use under this specification is on indoor concrete floors, particularly concrete basement floors, it can also be used on the interior surfaces of masonry walls and steel, wherever an acid and alkali-resistant coating is wanted. The paint dries mainly by evaporation.

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

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U. S. GOVERNMENT PRINTING OFFICE : 1974 O - 544-609/1597

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